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The exercise-hypogonadal male condition and endurance exercise training

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Abstract

An increasing number of research studies in men indicate endurance exercise training has significant effects upon the major male reproductive hormone, testosterone, and the hypothalamic-pituitary-testicular axis that regulates reproductive hormone production. This review article addresses a relatively new reproductive endocrine dysfunction found in exercising men, what has been deemed the “exercise-hypogonadal male condition”. Specifically, men with this condition exhibit basal (resting-state) free and total testosterone levels that are significantly and persistently reduced. The exact physiological mechanism inducing the reduction of testosterone is currently unclear, but is postulated to be a dysfunction (or perhaps a readjustment) within the hypothalamic-pituitary-testicular regulatory axis. The time course for the development of the exercise-hypogonadal condition or the threshold of exercise training necessary to induce the condition remains unresolved. The potential exists for these reduced testosterone levels within the exercise-hypogonadal male to be disruptive and detrimental to some anabolic or androgenic testosterone-dependent physiological processes. Regrettably, few research studies have addressed whether such processes are affected; thus, findings are equivocal. Conversely, the alterations in circulating testosterone brought about by endurance exercise training have the potential for cardiovascular protective effects and could be beneficial to the health of these men. Present evidence suggests this condition is limited to men who have been persistently involved in chronic endurance exercise training for extended periods of time (*i.e.*, years) and is not a prevalent occurrence. Nonetheless, many questions regarding the male reproductive endocrine adaptive process to exercise training still remain unanswered, necessitating the need for much further research.

Keywords

Endocrine; androgens; physical activity; stress

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Introduction

Athletes who participate in endurance events perform a tremendous amount of exercise training (1, 2, 3, 4). For example, it is not uncommon for a marathon runner to perform 150 to 200 kilometers of intensive running per week as part of their regular training. Exercise training to this extent results in positive physiological adaptations that are highly beneficial to the human body. There is an enhancement of the maximal cardiac stroke volume, maximal cardiac output, maximal arterial-venous oxygen differential, increased erythrocyte number, decreased levels of stored adiposity and increased skeletal muscle mitochondrial density (3). Such changes in these physiological parameters result in an increased human performance capacity. Yet exercise training to this extent can also place an incredible amount of stress and strain on the athlete's body and result in unwanted physiological responses and health problems. Development of the "Overtraining Syndrome" condition, which can completely compromise the ability of an athlete to perform (2), or musculo-skeletal trauma (*i.e.*, "pulled muscles") are common occurrences in athletes performing training at these high volumes.

One of the body's physiological systems that is extremely sensitive to the stress of exercise training is the endocrine system. This is in particularly true for the components of the endocrine system associated with the control and regulation of the reproductive system. During the last several decades, an increasing number of research studies point to how chronic exposure to endurance exercise training results in the development of a dysfunction within the reproductive components of the endocrine system of humans (5, 6, 7, 8, 9). The majority of the research on this topic has concentrated upon women athletes (10, 11, 12). However, recently investigators have begun to address the question of how exercise training affects the reproductive endocrine system in men too. Unfortunately, the number of findings on this latter topic is still relatively sparse as compared to the number of women-based studies.

Research on men shows the existence of a select group who through their exposure to chronic endurance exercise training have developed alterations in their reproductive hormonal profile - principally, low resting testosterone levels. The majority of these men display clinically "normal" levels of the hormone testosterone, but the levels are at the very low end of normal and in some cases reach a sub-clinical status. The consequences in these men of such hormonal changes are spermatogenesis and male infertility problems. The prevalence of the problems seems low, but as noted the research studies examining this condition and its consequences are few in the literature (13).

The terminology to refer to these "endurance trained men with low resting testosterone" has not been universally standardized. In 2004 researchers from our laboratory proposed use of the phrase the "exercise-hypogonadal male" as a name to refer to this condition (14, 15). This is the terminology we have chosen to use in our discussion throughout this paper. Men with this condition have certain characteristics and traits in common. These are summarized in Table 1.

This short review article presents an overview discussion of the major research literature addressing “exercise-hypogonadal males” and the physiological characteristics associated with the condition. Specifically, we address how endurance exercise training affects the male reproductive endocrine system to induce hypogonadal-like conditions resulting in suppressed circulating testosterone and in turn bringing about the exercise-hypogonadal condition. This article is intended to be an introduction to this topic. If the reader wishes a more detailed presentation on the topic they are directed to previously published work by Hackney and Dobridge (13).

Resting Responses

Retrospective comparative studies have reported significantly lower testosterone levels in exercise-hypogonadal men. In these studies, a single isolated basal blood sample (usually morning) was obtained and analyzed. The subjects of these studies have been almost exclusively distance runners who had been involved with exercise training in their sport for 1 to 15 years. Testosterone levels in the exercise-hypogonadal men were only 50–85% of the levels found in the control subjects (matched, sedentary) of these studies (7, 17, 18, 19, 20, 21). Research by Gullledge and Hackney found these low basal testosterone findings are highly reproducible and not just an aberration of the athletes’ seasonal training regime (22).

Research studies using a prospective approach also exist which support these findings. In these studies basal blood samples were collected over weeks or months while exposing subjects to endurance exercise training regimens. Yet, the findings from the prospective studies have not been universal or consistent. Several studies have revealed significant reductions (decreases of 20 – 40% from pre-training levels) in basal testosterone levels following 1 to 6 months of intensive training (22, 23, 24, 25, 26). Other studies, however, report no significant change in basal testosterone levels after 2 to 9 months of training (27, 28, 29, 30, 31, 32). Differences in the initial training status of the subjects, or the training dosage administered within the individual studies may perhaps account for the discrepancy in findings. It is also likely these studies have been conducted for too short a period of time (*i.e.*, months, whereas in the retrospective studies the men with low testosterone level have been training for years).

Exercise-hypogonadal men also display other reproductive hormonal abnormalities. The most frequently reported involves the lack of a significant elevation in basal luteinizing hormone in correspondence with the significant decreases in testosterone (*i.e.*, hypogonadotrophic-hypogonadism characteristics) (17, 21, 33). Additionally, it has been reported that resting levels of prolactin are decreased in these men (17, 21). These findings of altered basal prolactin and luteinizing hormone levels in exercise-hypogonadal men have been interpreted by some researchers as indicative of a dysfunction within the hypothalamic-pituitary-testicular axis. These prolactin and luteinizing hormone findings have been reported in several retrospective and prospective type studies (7, 17, 18, 21, 33, 34).

There are also retrospective investigations where basal blood specimens were collected more frequently, every 20 or 30 min for 4 to 8 hour periods, during the same day in exercise-hypogonadal men and compared to control subjects. Results are nearly identical to those of

the single blood sampling studies; that is, basal testosterone levels of the trained subjects were typically only 60 – 80% of those found in control sedentary men (18, 35). Basal lutropin levels were not significantly elevated in these frequent blood sampling studies either; again suggesting a possible dysfunction in the hypothalamic-pituitary-testicular regulatory axis.

The hormonal findings noted above have been found primarily in athletes involved in endurance exercise training (for example, marathon distance runners and tri-athletes). It is highly unlikely though that these hormonal alterations are limited to this athletic - exercise group alone. The prevalence of these phenomena in endurance athletes most likely represents the tendency of researchers to center their attention on this group, following the lead of early studies conducted on this topic. As exercise scientists expand their endocrinological studies to include other athletic - exercise groups involved with endurance training, it is highly likely that comparable data will come forward.

Exercise Responses

Few studies have evaluated how exercise-hypogonadal men and comparable sedentary men respond to exercise. Evidence suggests that in response to an exercise bout (high-intensity, maximal or sub-maximal in nature) the directions of the hormonal changes are similar; although, the absolute and relative magnitude of the hormonal responses varies greatly between each group of men. Testosterone, prolactin and cortisol are increased while the gonadotrophins are variable in both groups of men (17, 34). In the recovery from exercise the two groups differ distinctly. In sedentary men, the reproductive hormones display a normal negative feedback loop rebound inhibition during the hours after exercise (17, 36). Exercise-hypogonadal men, exhibit only a partial rebound effect or it is eliminated altogether. It is unclear whether this represents an adaptive adjustment in the sensitivity of the regulatory axis, or is reflective of a dysfunction in the axis. Most investigators have referred to such findings as dysfunctional in nature but more research work is necessary to determine this point exactly (17, 34, 36).

Studies Examining Mechanisms

Several research studies have gone beyond describing the existence of the exercise-hypogonadal condition and attempted to elucidate the mechanism of the proposed hypothalamic-pituitary-testicular axis dysfunction that exist in these men. These studies have focused on examining whether the dysfunction is at the hypothalamic or pituitary (central level) or at the testis (peripheral level) of the regulatory axis components.

Central mechanism research has seemed to focus upon alterations in prolactin and lutropin hormonal production and secretion. Such alterations in prolactin and lutropin production-secretion have been an area of extensive research in exercising women who develop reproductive dysfunctions (*i.e.*, amenorrhea). That is, the male-based research is being modeled after that which has been done on females (10, 37, 38).

An attenuated release of lutropin following injections of gonadotrophin-releasing hormone (GnRH [also called Gonadoliberin]) has been detected in these exercise-hypogonadal men

with low testosterone (31, 33, 39, 40). This suggests that perhaps there might be the development of some type of GnRH resistance or sensitivity at the anterior pituitary. Interestingly, results are ambiguous as to whether lutropin pulsatile characteristics (*i.e.*, pulse frequency and amplitude) are consistently affected in exercise-hypogonadal men or whether when such characteristics are effected if these in turn have any affect on testosterone production (18, 24, 35, 41, 42).

On the other hand, a greatly enhanced prolactin release in response to an exogenous stimulus of drugs or synthetic hormones has also been found in exercise-hypogonadal men. The hormone prolactin presents an interesting paradox in reproductive physiological function. Small amounts of prolactin seem necessary to work synergistically at the testis with lutropin, while excessive circulating levels disrupt both central and peripheral aspects of the hypothalamic-pituitary-testicular axis (41, 42).

Peripheral mechanism research has centered on alterations in the sensitivity of the testes to lutropin changes. Evaluations of testicular ability to produce and secrete testosterone in exercise-hypogonadal men have been contradictory. Some research suggests testicular steroidogenesis of testosterone is normal (33, 39), while some suggest it is impaired (40, 43, 44).

In a related manner, research shows acute pharmacological or pathological increases in cortisol and prolactin are associated with decreased circulating testosterone (41, 45, 46). It has been hypothesized that these hormonal changes serve as a potential mechanism for the low testosterone in exercise-hypogonadal men (6, 47, 45, 46). A single, acute exercise bout at high intensity (60% or greater of an individuals maximal aerobic capacity) can produce large transient increases in circulating prolactin and cortisol. These hormonal increases in turn could bring about the observed reductions in testosterone via the inhibitory actions of each of these hormones (47). New exercise work from Daly et al. supports this concept, with respect to cortisol, but much further research is necessary before definitive cause and effect conclusions can be drawn (48).

Exercise-Hypogonadal Condition - Consequences

Evidence suggests that the development of low resting testosterone levels in men doing endurance training has some detrimental effects on the testosterone-dependent physiological processes of the body. Hypothetically there is the potential for disruption in any of the processes influenced by testosterone. To date only a few studies have systematically studied this issue, and currently there are only a few reports of problems existing. For example, decreased spermatogenesis or oligospermic conditions have been found in some exercise-hypogonadal men (49, 50, 51, 52, 53). One well-controlled study on this topic was performed by Arce et al. (51). These investigators demonstrated that significant spermatogenesis problems existed in exercise-hypogonadal men. These findings though are not universal in the literature (30). Additionally, a lowered sex drive was found in some exercise-hypogonadal men, but a direct cause-and-effect linkage between a lower sex drive and lower testosterone was not demonstrated (43, 52, 54, 55, 56, 57). Other facts may be

affecting the libido status of these men (*eg.*, overall fatigue and psychological stress), more research is necessary on this topic (54, 58).

The other major androgenic-anabolic actions of testosterone are apparently not detrimentally affected in exercise-hypogonadal men. That is, thus far there is no evidence of decreased protein synthesis and muscle mass development in these men. Concern has been raised by some investigators about the impact of low testosterone levels on bone demineralization in exercise-hypogonadal men [again, modeled after the findings in exercising women (59)]. Several compelling case study reports have found exercise-hypogonadal men to have excessively low bone mineral density levels (60, 61). But, no conclusive experimental study findings currently supporting these findings have been published (62, 63).

Contrary to the above discussion, there could be beneficial physiological effects from the lowered testosterone associated with the condition. For example, some research indicates that lowering testosterone may have cardiovascular protective effects and decrease the risk of coronary heart disease (64). Investigators in Germany have provided strong evidence supporting this claim (65). Their research demonstrated that pharmacologically-induced reduction in endogenous testosterone levels results in significant increases in high-density lipoprotein in men. Whether the lowering of testosterone in exercise-hypogonadal men produces the same effect remains to be determined; but, most certainly presents an interesting point future researchers should address.

Summary and Conclusions

An increasing number of research studies in men indicate exercise training has significant effects upon the major male reproductive hormone, testosterone, and the hypothalamic-pituitary-testicular axis that regulates reproductive hormone levels. Specifically, findings suggest basal, resting-state testosterone (free and total levels) is significantly and persistently reduced in men involved in chronic endurance exercise training. This condition has been referred to as exercise-hypogonadism. The exact physiological mechanism inducing the reduction of testosterone is currently unclear, but is postulated to be a dysfunction within the hypothalamic-pituitary-testicular regulatory axis. It may also perhaps reflect a lowering of the set-point (*i.e.*, readjustment) of the axis for what is deemed a necessary minimal amount of circulating testosterone for proper physiological function. The time course for the development of the exercise-hypogonadal condition or the threshold of exercise training necessary to induce the condition remains unresolved. Much further scientific work is necessary to address these latter points (66).

The potential exists for the reduced testosterone of exercise-hypogonadal men to disrupt some anabolic or androgenic testosterone-dependent physiological processes. Unfortunately a very limited number of studies have addressed whether such processes are affected, and thus findings are inconclusive on this issue. Conversely, the alterations in testosterone levels brought about by endurance training could have cardiovascular protective effects and thus be beneficial to the health of these men.

During the last 30 years a large number of research studies have been conducted examining the reproductive endocrine dysfunction in exercising women. The total number of similar studies examining men is still relatively small and thus many questions regarding the male reproductive endocrine adaptive process to exercise training remain. Thus, there is a great need for continued investigation in this area of endocrinological research.

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TABLE 1.

The table contains the common characteristics and traits of men displaying the exercise-hypogonadal condition.

(i)	They have low resting basal testosterone levels, typically 50–75% that of normal, healthy, age-matched sedentary men.
(ii)	Their low testosterone levels do not appear to be a transient phenomenon related to the acute stress-strain of exercise (16).
(iii)	In many cases, it appears an adjustment in the regulatory axis (to allow a new lower set-point for circulating testosterone) has occurred (16).
(iv)	They typically have a history of early involvement in organized sport and exercise training. This has resulted in these men having many years of almost daily exposure physical activity.
(v)	The type of exercise training history most frequently seen in these men is prolonged, endurance-based activities such as: distance running, cycling, race walking, and the triathlon training.